HARBORTECH SOLUTIONS

Cloud Operations Support Practice

IT 4560 · Week 06 Lab Submission

Do We Need a Server? Compute Model Recommendation

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SECTION 2 | EVIDENCE REVIEW AND ANALYSIS

Response 1 | Daily report: compute model and workload evidence

I would evaluate AWS Lambda first for the daily reporting task because it only runs once each morning and then stops. One piece of evidence is that the task completes quickly and does not need to run continuously. Another is that it keeps no local state between runs, which makes it a good candidate for event-driven serverless computing. Lambda could run the report when scheduled without requiring an EC2 instance to stay running all day. This would reduce server-management responsibilities and potentially lower costs. HarborTech would still be responsible for maintaining the code, testing the report, managing permissions, and monitoring for errors.

Response 2 | API Gateway and Lambda: request flow and responsibilities

AWS Lambda and API Gateway could work together to support the small REST API. API Gateway would receive incoming HTTP requests and route them to the appropriate Lambda function. Lambda would execute the application code and return a response without HarborTech having to manage the underlying servers. This design could be useful because the API receives intermittent requests instead of requiring constant server activity. However, HarborTech would still be responsible for application logic, validating input, handling errors, securing access, and maintaining any connections to databases or other services. The team would also need to test performance and monitor the API to make sure it meets its requirements.

Response 3 | Continuously running workload: VM, container, or Lambda

Lambda would be a weaker choice for this application because it runs continuously and requires a custom operating-system package and host-level troubleshooting access. Lambda limits control over the underlying operating system and is designed for function-based execution rather than managing a persistent host environment. A virtual machine may be more appropriate if the application needs direct operating-system access, custom system configurations, or extensive host-level troubleshooting. A container could also work if the application can run with its dependencies packaged together and does not require control of the host operating system. Before choosing between a VM and a container, I would gather more evidence about the required operating-system package, whether the application needs a persistent process, how much host access is necessary, the expected workload, and the team's container-management capabilities. This evidence would help determine which option provides the needed control without adding unnecessary operational complexity.

SECTION 3 | COMPUTE MODEL RECOMMENDATION

HarborTech internal recommendation | TKT-2026-0006

HarborTech Compute Model Recommendation
Ticket:TKT-2026-0006
Prepared by:Junior Cloud Operations Intern
Subject: Compute Model Evaluation for Reporting Task and REST API

1. Current Server-Based Condition
Both workloads currently share one EC2 instance. The reporting task runs once each morning and stops after completing its work, while the small REST API receives intermittent requests throughout the day. The EC2 instance remains lightly utilized for long periods, but HarborTech and the client still carry operational responsibilities for the guest operating system, application runtime, security updates, and troubleshooting. Based on the available evidence, evaluating alternative compute models could reduce unnecessary server-management work.

2. Workload Evidence and Recommendations

Workload A: Daily Reporting Task

Evidence: The reporting task runs once each morning, completes quickly, does not maintain local state between runs, and leaves the EC2 instance idle for most of the day.

Recommendation:Evaluate AWS Lambda as the preferred compute model. Lambda can execute the reporting code on a schedule without requiring a dedicated EC2 instance to remain running solely for this task.

Operational responsibility tradeoff:AWS would manage the underlying execution infrastructure, reducing HarborTech's responsibility for server provisioning and guest operating system maintenance for this workload. HarborTech would still maintain the reporting code, configure the schedule and permissions, validate report accuracy, monitor failures, and manage access to any required data sources.

Workload B: Small REST API

Evidence:The API receives intermittent requests throughout the day, and the shared EC2 instance remains lightly utilized for extended periods.

Recommendation:Evaluate Amazon API Gateway with AWS Lambda as a serverless architecture. API Gateway could receive HTTP requests and route them to the appropriate Lambda function, while Lambda executes the application logic and returns responses. This could reduce the need to operate a continuously available application server.

Operational responsibility tradeoff: AWS would manage the underlying server infrastructure for API Gateway and Lambda. HarborTech would remain responsible for API design, application code, authentication and authorization configuration, input validation, error handling, logging, monitoring, and database or service integrations. The team would also need to verify that the design meets the API's latency, availability, security, and performance requirements.

3. VM and Container Considerations

A virtual machine remains appropriate for workloads that require guest operating system control, custom system-level packages, persistent processes, or host-level troubleshooting. Containers may be appropriate when application dependencies need consistent packaging and isolation but the workload can share a host operating system kernel.

Neither a VM nor a container should be selected automatically simply because the workloads currently run on EC2. The choice should be based on actual operating system requirements, runtime behavior, deployment needs, and the operational capabilities of the team. For the two workloads in this ticket, serverless is the initial recommendation to evaluate, subject to validation.

4. Validation Evidence to Collect

Before recommending implementation, HarborTech should collect and review the following evidence:

- Reporting task:Actual execution duration, schedule requirements, memory and CPU needs, local file dependencies, data access requirements, and whether the task can run successfully in Lambda's execution environment.
-REST API: Request volume and traffic patterns, response-time requirements, authentication needs, concurrent request levels, payload sizes, database connections, and availability requirements.
-Current EC2 environment: Instance utilization, application and operating system dependencies, current hosting costs, existing monitoring data, and whether other services depend on the shared instance.

-Migration testing:Successful report execution, API functional and load tests, permission checks, error handling, logging, and rollback procedures in an approved test environment.
-Cost and service limits: Estimated costs for the proposed services, expected invocation volume, execution duration, data transfer, API Gateway usage, and any applicable service quotas.

The current evidence supports evaluating serverless options, but it does not establish that either migration has been tested or approved.

5. Assumptions and Required Escalation

The recommendation assumes that the reporting task can execute within Lambda's runtime and resource limits and that the API does not require a persistent process or unsupported host-level access. These assumptions must be validated before selecting the final architecture.

The following decisions should be escalated to the authorized cloud architect or project owner before implementation:

1. Approval to migrate either workload away from the shared EC2 instance.
2. Confirmation of the target architecture, including Lambda, API Gateway, and any required supporting services.
3. Approval of IAM permissions, security controls, network connectivity, and access to data sources.
4. Confirmation of cost estimates, performance requirements, service quotas, and rollback plans.
5. Approval of the migration schedule and any changes that could affect production availability.

As a junior cloud operations intern, I would document the findings, gather the required evidence, and submit the recommendation for review rather than making unapproved infrastructure or permission changes.

6. Final Recommendation
HarborTech should evaluate AWS Lambda first for the daily reporting task and evaluate API Gateway with Lambda for the intermittent REST API. These options could reduce the operational burden associated with maintaining a shared EC2 instance for workloads that do not appear to require continuous server operation. A VM or container should remain an option if validation identifies persistent execution, custom operating system, or host-access requirements that make serverless unsuitable.

The final decision should be based on measured workload behavior, successful testing, security and cost reviews, and approval from the authorized decision-maker. No production migration should occur until these checks and approvals are complete.

SECTION 4 | HARBORTECH OPERATIONS PLAYBOOK

Direct GitHub URL | week06-containers-serverless.md

https://github.com/Kiyahbee205/IV-week6.0/upload/main

PLAYBOOK CONFIRMATION

Aakiyah confirms that week06-containers-serverless.md has been published to a public GitHub repository and is accessible at the URL above.